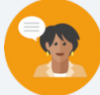


9.6 Finding Volume

Lesson Introduction

 Benchmark	MA.912.GR.4.5 Solve mathematical and real-world problems involving the volume of three-dimensional figures limited to cylinders, pyramids, prisms, cones, and spheres.		 Learning Target	I can find the volume of spheres, prisms, cones, pyramids, and cylinders.
 Benchmark Coherence	Building On			Working Toward
	MA.6.GR.2.3 Solve mathematical and real-world problems involving the volumes of right rectangular prisms with positive rational-number edge lengths using a visual model and a formula. MA.7.GR.2.3 Solve mathematical and real-world problems involving volumes of right circular cylinders.			N/A
 Essential Questions	- How can I find the volume of spheres, prisms, cones, pyramids, and cylinders? - What are the formulas for the volume of spheres, prisms, cones, pyramids, and cylinders? - How can I derive the formulas for the volume of spheres, prisms, cones, pyramids, and cylinders?			
 Lesson Narrative	This lesson guides students to develop a conceptual understanding of the formulas for finding volume. The collaboration in 9.6.2 Guided Instruction provides students the opportunity to discuss the mathematics and discover the relationships between the volumes of the cylinder, cone, pyramid, rectangular prism, and the sphere. This lesson also introduces the concept of composite figures and how to find their volumes.			
 Component Summary	9.6.1 Warm-Up (~5 min.)	Choosing between two pools based on dimensions (<i>SE p. 83</i>)	9.6.3 Guided Practice (5-10 min.)	Finding the volume of a hemisphere and a composite shape (<i>SE p. 87</i>)
	9.6.2 Guided Instruction (15-20 min.)	Finding volume of spheres, prisms, cones, pyramids, cylinders, and composite figures (<i>SE p. 83</i>)	9.6.4 Your Turn! (5-10 min.)	Finding the volume of a cylinder and a pyramid (<i>SE p. 87</i>)
	9.6.5 Wrap-Up (~5 min.)	Finding the volume of a composite figure (<i>SE p. 88</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Cone - Pyramid - Sphere <u>Additional Terms:</u> - Volume - Prism - Cylinder		(Optional) Water (Optional) Rice, Sugar, or Sand	If available, find a 3D model of a sphere, prism, cone, pyramid, and cylinder.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may divide the radius by 2 instead of dividing the volume by 2 to find the volume of a hemisphere. - Students may use the wrong formulas to find the volume of the individual three-dimensional figures (i.e. students may confuse the volume formulas for cylinders and cones). - Students may substitute in the diameter measure instead of the radius measure into a formula. - Students may not find the individual volumes of the figures that make up a composite figure.	- Consider having students make a formula chart to reference the formulas after students have made connections and derived the formulas. - If 3D models are not available, consider using an online visual such as GeoGebra to model the three-dimensional figures.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder The 9.6.2 Guided Instruction discovery portion of the lesson lends itself well to noticing and wondering, especially relating to the formulas for volume. Allow students the opportunity to notice and wonder throughout the lesson, as this should better help students conceptualize the formulas for volume.	MA.K12.MTR.6.1: Reasonableness Students will work through a series of questions to anticipate how the volume of a cone relates to a cylinder and the volume of a pyramid relates to a prism. Students then use the formulas to determine the volumes of figures. Students should evaluate their results, asking, “Does this solution make sense?” to monitor the reasonableness of their solutions.
 Differentiate Instruction	Consider using visuals for students to conceptualize the concept of volume. There are many examples of applets on GeoGebra that would help students to better understand how volume formulas are derived. This will provide another entry-point for students to derive the formulas for the volumes of various three-dimensional figures.	
Lesson Notes:		

9.6.1 Warm-Up: Would You Rather?

Component Overview: Students begin by deciding which of the two pools they would want to have. Students are to justify their answers using mathematics.



9.6.1 Warm-Up: Would You Rather?

1. Would you rather have a pool that is 20 meters (m) by 8 m by 6 m, or a pool that is 1550 centimeters (cm) by 650 cm by 725 cm?

- a. Use mathematics to compare the two pools.

Pool 1 volume: $20 \times 8 \times 6 = 960 \text{ m}^3$

Pool 2 volume: $15.5 \times 6.5 \times 7.25 = 730 \text{ m}^3$

- b. Which pool would you rather have?

Answers will vary. I would rather have pool 1 as it is larger.



To effectively compare the sizes of the two pools, students must convert centimeters to meters.



For a kinesthetic approach for 9.6.1 Warm-Up, poll the class and see how many students chose which option. Polling could involve a simple thumbs-up.

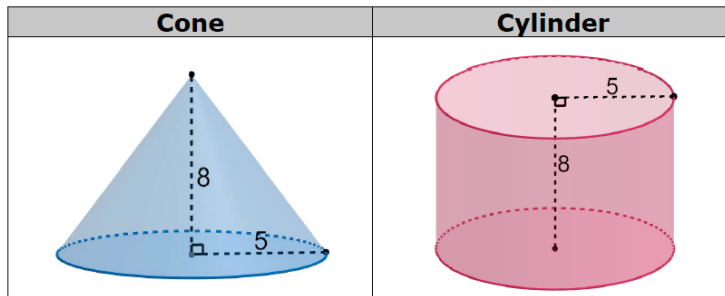
9.6.2 Guided Instruction: Finding Volume

Component Overview: Students will begin working with the sphere, cone, cylinder, prism, and pyramid and derive formulas for finding the volume of each. The purpose of this component is to conceptualize the volume formulas by showing the similarities between the cone and the cylinder as well as similarities between the rectangular prism and the pyramid. Composite figures are also briefly introduced.



9.6.2 Guided Instruction: Finding Volume

1. A cone and cylinder are shown, where the radius of the base of each is 5 units and the height of each is 8 units.



- a. Suppose the cone is filled with water. Discuss with your partner how many cones you think you would need to fill the cylinder with water. Write your estimate.
Answers will vary. I think that there would be 3 cones worth of water to fill the cylinder.
- b. Given the volume of a cylinder is $V = Bh$, discuss with your partner how to rewrite the formula to find the volume of the cone.
Answers will vary. I know it would need to involve pi and a fraction – it is smaller than the cylinder.
- c. The volume formula for a cone is $V = \pi r^2 \frac{h}{3}$



A **cone** is a three-dimensional figure with a circular base and an apex that is connected to the base by a collection of line segments that form a curved surface.



To better prompt student thinking about the formulas of the cone and the cylinder, consider using the following questions:

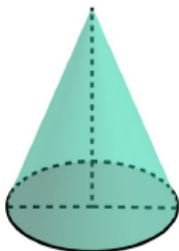
- What are the main differences between the cone and the cylinder?
- How are the cone and the cylinder similar?
- How can the similarities of the cone and cylinder help to develop a formula to find the volume?



For a kinesthetic approach for this question, consider having a 3D model of a cylinder and of a cone. Pour either water, sugar, sand, or rice from the cone into the cylinder to show that it takes the volume of three cones to make a cylinder.

9.6.2 Guided Instruction: Finding Volume (continued)

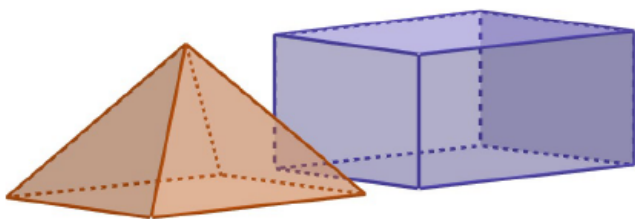
2. A cone is shown with a diameter of 16 millimeters (mm), a height of 22 mm, and a slant height of 23.4 mm.



Determine the volume of the cone, in terms of π .

$$\begin{aligned} V &= \pi r^2 \frac{h}{3} \\ &= \pi 8^2 \frac{22}{3} \\ &= \pi (64) \left(\frac{22}{3} \right) \\ &= \frac{1408}{3} \pi \text{ mm}^3 \end{aligned}$$

3. The bases of the rectangular pyramid and rectangular prism shown have congruent bases, and each has the same height.



- a. Suppose the rectangular pyramid is filled with water. Discuss with your partner how many rectangular pyramids you think you would need to fill the rectangular prism with water. Write your estimate.

Answers will vary. I believe that 4 rectangular pyramids would fill the rectangular prism.



For a visual approach to this activity, consider the use of digital applets for making sense of how many pyramids fill a rectangular prism.



What are the main differences between the rectangular prism and the pyramid?

9.6.2 Guided Instruction: Finding Volume (continued)

- b. Given the volume of a prism is $V = B \cdot h$, discuss with your partner how to rewrite the formula to find the volume of the pyramid.

Answers will vary. I believe that the formula might be $v = \frac{1}{2}b \cdot h$

- c. The volume formula for a regular pyramid is

$$V = \frac{l \cdot w \cdot h}{3}$$



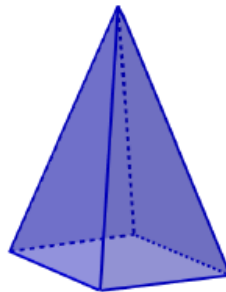
A **pyramid** is a figure with a polygonal base and triangular faces. The triangular faces have the same size and shape and they connect the sides of the base to a common point called the apex.



Common Misconception

Students may forget to put radius into the formula when given diameter. Remind students of this when solving to find volumes of spheres.

4. A pyramid is shown, whose base is a rectangle with dimensions of 10 centimeters (cm) by 14 cm. The height of the pyramid is 17 cm. Determine the volume of the pyramid.



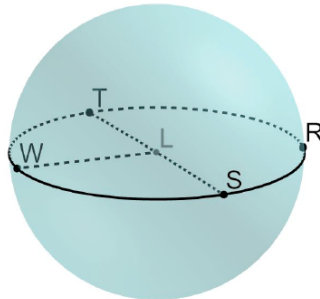
$$\begin{aligned} V &= \frac{10 \cdot 14 \cdot 17}{3} \\ &= \frac{2,380}{3} \text{ cm}^3 \end{aligned}$$

9.6.2 Guided Instruction: Finding Volume (continued)



A **sphere** is a three-dimensional figure, where all points are equidistant from a point called the center.

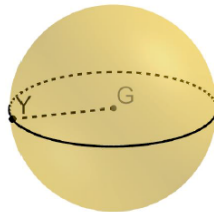
A sphere with center L is shown, where $\overline{LW}$ is a radius and $\overline{TS}$ is a diameter. The circle that passes through the points $TRSW$ is known as a great circle since it is the largest possible circle that can be made around the sphere.



The volume formula for a sphere is $V = \frac{4}{3}\pi r^3$.

5. Find the volume of the sphere G , where $GY = 3$ using $\pi \approx 3.14$.

$$\begin{aligned} V &= \frac{4}{3}\pi(3^3) \\ &= \frac{4 \cdot 3.14 \cdot 27}{3} \\ &= \frac{339.12}{3} \\ &= 113.04 \text{ units}^3 \end{aligned}$$



Use the following questions to guide student thinking on question 6:

- What common features do the cone and cylinder share? (Same base.)
- Would the area of the bases of the cylinder be equal to the area of the base of the cone? (Yes.)
- How do you know that the diameters, radii, and the bases all have the same area in the decomposed shapes of the composite figure?



Common Misconception

Students may forget to find the volume of both individual shapes and then add them together. Have students find the volume of each shape individually and label.



For a visual approach to these questions, have students make a table like the one shown below.

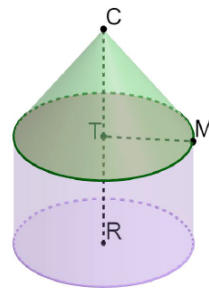
Three-Dimensional Figure	Work	Volume
Pyramid		
Rectangular Prism		
Composite Volume		

9.6.2 Guided Instruction: Finding Volume (continued)

6. Composite figures are made up of more than one figure. The composite figure shown is a cone on top of a cylinder.

Check all that must be true about the composite figure.

- ☐ The heights have the same measure.
- ☒ *The diameters have the same measure.*
- ☒ *The radii have the same measure.*
- ☒ *The bases have the same area.*
- ☐ The volume of this composite figure can be found using the formula $V = 2\pi r^2 h$.

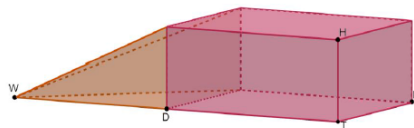


7. Discuss with your partner the statements that were false. Rewrite each false statement so that it would be true.

The heights must be labeled or be able to be solved for.

The volume of the composite figure can be found using the formula $V = \left(\pi r^2 \frac{h_{\text{cone}}}{3}\right) + (\pi r^2 h_{\text{cylinder}})$

8. A composite figure is shown, where $HT = 3.2$, $KT = 4.8$, $DT = 10.7$, and $WD = 11.1$. WD is the height of the pyramid. Find the volume of the composite figure.



$$\begin{aligned}
 V &= (l \cdot w \cdot h) + \left(\frac{l \cdot w \cdot h}{3}\right) \\
 &= (3.2 \cdot 4.8 \cdot 10.7) + \left(\frac{3.2 \cdot 11.1 \cdot 4.8}{3}\right) \\
 &= 164.352 + 56.832 \\
 &= 221.184 \text{ units}^3
 \end{aligned}$$

9.6.3 Guided Practice: Finding Volume

Component Overview: Students are moving to practicing without being given figures. Questions 1-2 require students to find the volume of a hemisphere. Have students be sure to find the volume of the sphere and then divide by 2.



9.6.3 Guided Practice: Finding Volume

- Find the volume of a hemisphere that has a diameter of 9.4 inches. Round your answer to the nearest thousandth.

$$\begin{aligned}
 V &= \frac{2}{3}\pi r^3 \\
 &= \frac{2}{3}\pi \cdot 4.7^3 \\
 &= \frac{207.646\pi}{3} \\
 &= 217.446 \text{ in}^3
 \end{aligned}$$

- A composite figure consists of a hemisphere and a cone. The base of the cone and the great circle of the hemisphere have the same diameter. The radius of the hemisphere is $4\frac{3}{4}$ centimeters (cm). The height of the cone is $9\frac{1}{12}$ centimeters (cm). Find the volume of the composite figure using $\frac{22}{7}$ for π . Round your answer to the nearest thousandth.

$$\begin{aligned}
 V &= \left(\frac{2}{3}\pi r^3\right) + \left(\pi r^2 \frac{h}{3}\right) \\
 &= \left(\frac{2}{3}\right)\left(\frac{19}{4} \cdot \frac{19}{4} \cdot \frac{19}{4}\right)\left(\frac{22}{7}\right) + \left(\frac{1}{3}\right)\left(\frac{109}{12}\right)\left(\frac{19}{4} \cdot \frac{19}{4}\right)\left(\frac{22}{7}\right) \\
 &= \left(\frac{2}{3}\right)\left(\frac{6859}{64}\right)\left(\frac{22}{7}\right) + \left(\frac{109}{36}\right)\left(\frac{361}{16}\right)\left(\frac{22}{7}\right) \\
 &= \frac{13,718}{192}\left(\frac{22}{7}\right) + \frac{39,349}{576}\left(\frac{22}{7}\right) \\
 &= \frac{75,449}{336} + \frac{432,839}{2016} \\
 &= \frac{885,533}{2016} = 439.252 \text{ cm}^3
 \end{aligned}$$



Common Misconception

Students may substitute in the diameter instead of the radius into the formula. Students may also forget to divide the volume of a sphere by 2 when finding the volume of a hemisphere.

9.6.4 Your Turn!

Component Overview: Students are asked to find the volumes of two figures without being given images of the figures.



9.6.4 Your Turn!

- Find the volume of a cylinder with a diameter of 18.3 millimeters (mm) and a height of 6.4 mm. Round your answer to the nearest thousandth.

$$\begin{aligned} V &= \pi r^2 h \\ &= \pi \cdot 9.15^2 \cdot 6.4 \\ &= 535.824\pi \\ &= 1,683.341 \text{ mm}^3 \end{aligned}$$

- Find the volume of a pyramid whose height is 15.7 inches (in.), and whose base is a rectangle with dimensions of 7.6 in. by 12.4 in. Round your answer to the nearest thousandth.

$$\begin{aligned} V &= \frac{lwh}{3} \\ &= \frac{7.6 \cdot 12.4 \cdot 15.7}{3} \\ &= \frac{1479.568}{3} \\ &= 493.189 \text{ in}^3 \end{aligned}$$



Use the following questions to guide students through a common misconception:

- What is the difference between the radius and diameter?
- How is radius found when given the diameter?
- How can you use the volume of a cylinder formula to find the volume now that the radius is given?

9.6.5 Wrap-Up: Ticket Out the Door

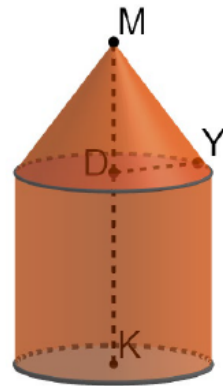
Component Overview: Students complete a question to measure mastery of the benchmark and the lesson target. Students should be allowed to access a calculator to complete 9.6.5 Wrap-Up.



9.6.5 Wrap-Up: Ticket Out the Door

- Find the volume of the composite figure if $MD = 17$, $DY = 8.5$, and $KD = 26$. Round your answer to the nearest hundredth.

$$\begin{aligned}
 V &= (\pi r^2 h) + \left(\pi r^2 \frac{h}{3}\right) \\
 &= (\pi 8.5^2 \cdot 26) + \left(\pi 8.5^2 \frac{17}{3}\right) \\
 &= (\pi 72.25 \cdot 26) + \left(\frac{\pi \cdot 72.25 \cdot 17}{3}\right) \\
 &= (5,901.482) + (1,286.220) \\
 &= 7,187.70 \text{ units}^3
 \end{aligned}$$



The information provided from the students here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.